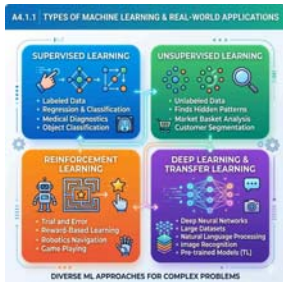


A 4.1.1 Describe the types of machine learning and their applications in the real world



A4.1.1 Describe the types of machine learning and their applications in the real world

Machine learning (ML) is a branch of artificial intelligence in which a system learns from data rather than following a fixed set of programmed rules. This page covers five approaches to machine learning, their key characteristics, and their real-world applications. Two of these approaches, Deep Learning and Transfer Learning, are not types of learning in the same sense as the others; understanding the distinction is important for exam answers.

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Key Questions

- What are the five approaches to machine learning covered in this topic, and what distinguishes each one?
- How do supervised and unsupervised learning differ in terms of the data they require and the tasks they perform?
- What is reinforcement learning, and why is it fundamentally different from the other two learning types?
- Why are deep learning and transfer learning described differently from supervised, unsupervised, and reinforcement learning?
- How would you identify which ML approach is most appropriate for a given real-world problem?

IB Command Terms

Describe

Give a detailed account of the features or characteristics of each approach. For each: state what it is, what kind of data it uses, what task it performs, and give a real-world application.

What to include

For each approach: (1) definition, (2) key characteristic such as data type or mechanism, (3) task type, (4) named real-world application with brief explanation. For deep learning and transfer learning, also note their relationship to the three fundamental types.

What is Machine Learning?

Traditional programming requires a developer to write explicit rules for every situation the system might encounter. This is practical for simple, well-defined problems but fails when the rules are too complex or too numerous to specify manually.

Machine learning takes a different approach. Instead of writing rules, a developer provides a large amount of data and a learning algorithm. The algorithm identifies patterns in that data and builds a model that can apply those patterns to new, unseen examples. The rules are learned from data rather than written by hand.

TRADITIONAL PROGRAMMING VS MACHINE LEARNING

Traditional programming: input data + explicit rules = output.

Example: IF email contains 'free prize' THEN mark as spam. This breaks whenever spam changes its wording.

Machine learning: input data + desired outputs (or interaction) = learned model.

Example: show the system 50,000 labelled emails; it learns the features that distinguish spam from legitimate email and applies that to new messages it has never encountered.

The five approaches to machine learning

The five approaches below are not all equivalent categories. Supervised learning, unsupervised learning, and reinforcement learning are the three fundamental paradigms. They describe how the learning process is structured. Deep learning is a technique that can be applied within any of those

paradigms. Transfer learning is a strategy for reusing an existing model. This distinction is important in exam questions.

1. Supervised Learning - *Uses labelled data to predict outcomes*

—

A hospital has 20,000 chest X-rays. Each one has already been examined and labelled by a radiologist as 'tumour present' or 'no tumour'. How could a computer use these labelled examples to assess future X-rays it has never seen?

Supervised learning trains a model on a dataset in which every example already has a correct answer attached. This correct answer is called a label. The model learns the relationship between the input data and its label, then uses that relationship to predict labels for new, unlabelled data.

Key characteristic	Every training example has a label -- a known correct answer -- attached by a human or an existing system.
Task type: Classification	The model predicts a category. Output is one of a fixed set of classes. Examples: spam or not spam (two classes); classifying handwritten digits 0 to 9 (ten classes).
Task type: Regression	The model predicts a numerical value on a continuous scale. Examples: predicting a patient's blood pressure given their health metrics; estimating a property's sale price given its features.
Limitation	Producing labelled training data requires human effort. For large or specialist datasets this is expensive and time-consuming.

Analogy A student learns to identify bird species by studying a field guide in which every photograph is labelled with the species name. After enough examples, they can identify species they have not seen before. The photographs are the inputs; the species names are the labels.

Real-World Applications

- **Sentiment analysis:** A model trained on thousands of product reviews labelled 'positive', 'neutral', or 'negative' by human annotators can then classify new reviews automatically.

- **Medical imaging diagnostics:** A model trained on thousands of scans already diagnosed by specialist radiologists learns to detect tumours, fractures, or abnormalities in new scans.
- **Spam detection:** A model trained on millions of emails labelled spam or not spam learns which features - subject line patterns, sender behaviour, link density - predict spam.

Common Exam Mistake Students often select supervised learning for any task that involves prediction or a 'correct answer'. The critical test is whether labelled training data exists. If the data has no pre-existing labels, supervised learning cannot be used.

2. Unsupervised Learning - *Finds hidden patterns in unlabelled data*

A supermarket has five years of transaction records - every purchase made by every customer. The data has no labels or categories attached. How could a computer begin to find meaningful structure in this data without being told what to look for?

Unsupervised learning trains on data with no labels and no correct answers. The algorithm must discover structure in the data entirely on its own. The two most common tasks are clustering - grouping similar items together -- and association - finding items that frequently occur together.

Key characteristic	Training data has no labels. The algorithm defines its own categories or finds its own patterns.
Task type: Clustering	The model groups data points that are similar to each other. It decides the groupings without being told what the groups should be. Example: grouping customers by purchasing behaviour without pre-defining customer types.
Task type: Association	The model identifies items or events that frequently appear together in the same record. Example: identifying that customers who buy bread also frequently buy butter in the same transaction. This is the basis of market basket analysis.
Limitation	The model cannot label or name the groups it discovers. Interpreting what a cluster represents requires human judgement.

Analogy A librarian is given 10,000 books with no genre labels and asked to organise them. Without any instructions, they begin sorting by subject matter, style, and audience -- creating their own categories. The categories are meaningful, but the librarian assigned the names, not the data.

Real-World Applications

- **Market basket analysis:** A supermarket analyses transaction records to discover which products are frequently bought together. This informs store layout, promotions, and online product recommendations.
- **Customer segmentation:** An online retailer groups customers by purchasing behaviour, browsing patterns, and demographics to target different groups with different marketing messages.
- **Anomaly detection:** A bank identifies unusual transaction patterns that differ significantly from a customer's normal behaviour, flagging potential fraud without having labelled fraud examples to train on.

Common Exam Mistake Unsupervised learning discovers groups in existing data - it does not predict which group a new data point belongs to. That second step, classifying new data using the discovered groups, requires a supervised model trained on the outputs of the unsupervised stage.

3. Reinforcement Learning - *Learns through trial, error and reward*

Think about how you learned to ride a bicycle. You were not given a dataset of correct pedalling instructions. What actually drove your learning? What happened when you fell, and what happened when you stayed upright?

Reinforcement learning works on a completely different principle from supervised and unsupervised learning. There is no dataset. Instead, an agent learns by interacting with an environment, taking actions, and receiving feedback in the form of rewards or penalties. Over many interactions, the agent learns a policy - a strategy for choosing actions that maximises its total reward over time.

Key characteristic	No static dataset. The agent generates its own experience by interacting with an environment.
Core concept: Agent	The learner or decision-maker. It observes the current state of the environment and selects an action.
Core concept: Environment	The world the agent interacts with. It responds to the agent's actions by transitioning to a new state and issuing a reward.
Core concept: Reward	A numerical signal indicating how good or bad the outcome of an action was. The agent's goal is to maximise cumulative reward over time.
Core concept: Policy	The strategy the agent has learned -- a mapping from states to actions that the agent uses to decide what to do in any given situation.
Core loop	Agent observes state, selects action, receives reward and new state, updates policy. Repeat.

Analogy Training a dog to sit. The dog (agent) tries different behaviours (actions) in response to the command (state). A treat (positive reward) follows the correct behaviour; nothing follows an incorrect one. After many repetitions, the dog learns the policy: 'sit when I hear this command'. No labelled dataset was involved.

Real-World Applications

- **Robotics navigation:** A robot learns to move through a physical environment by receiving a positive reward for reaching a destination and a negative reward (penalty) for colliding with obstacles. It improves its movement policy through millions of simulated interactions.
- **Game playing:** An AI learns to play chess, Go, or video games by playing millions of games against itself or other agents, learning which sequences of moves lead to victory. This approach produced systems that play at a level beyond the best human experts.
- **Resource management:** Data centre cooling systems have been trained using RL to reduce energy consumption by learning policies that balance cooling efficiency against hardware safety.

Common Exam Mistake Reinforcement learning is confused with supervised learning because both involve a form of 'correct' outcome. The key difference is that in RL there is no labelled training dataset prepared in advance. The agent generates its own training experience through interaction. The reward signal is received after the action, not provided as a label before training.

4. Deep Learning (*Technique*) - Neural networks for complex pattern recognition —

Your phone can unlock by recognising your face. It was not programmed with a description of what your face looks like. The feature that makes your face unique - the distance between your eyes, the shape of your jaw - was never written into any rule. How do you think it learned?

Deep learning is not a separate category of learning alongside supervised, unsupervised, and reinforcement learning. It is a technique, a specific architectural approach, that can be applied within any of those paradigms. It uses Artificial Neural Networks (ANNs) with many layers, where each layer learns increasingly abstract representations of the input data.

Deep learning is especially effective with unstructured data such as images, audio, and text, where traditional ML algorithms require extensive manual feature engineering to produce useful inputs.

Deep learning learns its own features from raw data.

Category	Technique -- can be used within supervised, unsupervised, or reinforcement learning.
Key characteristic	Multi-layered Artificial Neural Networks (ANNs). Each layer transforms the input into a more abstract representation.
Strength	Unstructured data (images, audio, text, video) where useful features are not obvious and must be learned from raw input.
Limitation	Requires large amounts of training data and significant computing power. The learned representations are difficult to interpret (often called a 'black box').

Analogy When learning to recognise a cat, the human visual system does not apply a list of rules. It builds up layers of recognition: first detecting edges, then shapes, then combinations of shapes that form ears, eyes, and whiskers, then the overall pattern 'cat'. Each layer processes the output of the layer below. Deep learning mimics this layered processing in software.

Real-World Applications

- **Object detection and classification:** Self-driving vehicles use deep learning models to identify pedestrians, cyclists, road signs, and other vehicles from camera and sensor data in real time.
- **Natural language processing (NLP):** Large language models, voice assistants, and machine translation services rely on deep learning to process and generate human language.
- **Medical image analysis:** Deep learning models trained on large sets of medical scans can detect cancerous lesions, retinal disease, and bone fractures with accuracy comparable to specialist clinicians.

Common Exam Mistake Deep learning is frequently described in student answers as a type of machine learning separate from and parallel to supervised and unsupervised learning. It is not. It is a technique it is a way of building models that can be used within those paradigms. An exam answer should make this explicit: 'Deep learning is a technique that uses multi-layered neural networks and can be applied to supervised, unsupervised, or reinforcement learning tasks.'

5. Transfer Learning (*Strategy*) - Reuses pre-trained models as a shortcut

If you already know how to play the piano, how long would it take you to learn the organ compared to someone who has never played any musical instrument? Which skills transfer across, and which do you have to learn from scratch?

Transfer learning is not a fundamental type of learning. It is a strategy for building new models efficiently by reusing knowledge encoded in an existing model. A model already trained on a large dataset for one task is taken as a starting point and then fine-tuned -- adjusted with a small amount of new data -- for a second, related task.

Fine-tuning means taking the pre-trained model's weights and continuing to train them on the new, smaller dataset. The model retains the general knowledge it acquired from the original large dataset and adapts it to the specifics of the new task.

Category

Strategy -- a way of applying deep learning models more efficiently.

Key characteristic	A pre-trained model from one task is used as the starting point for a second, related task.
Benefit	Requires far less labelled data and training time than building a new model from scratch. Particularly valuable when the target dataset is small or expensive to produce.
Most commonly used with	Deep learning models, because they encode rich, reusable representations of their input domain.
Fine-tuning defined	The process of taking a pre-trained model and continuing to train its parameters on new data for the target task. The model adapts without losing its general knowledge.

 **Analogy** A civil engineer who already understands structural loading, materials, and physics can learn to design a specific new type of bridge far faster than someone starting from no engineering background. They do not re-learn the fundamentals, they apply existing knowledge to new constraints. Transfer learning applies the same principle to machine learning models.

Real-World Applications

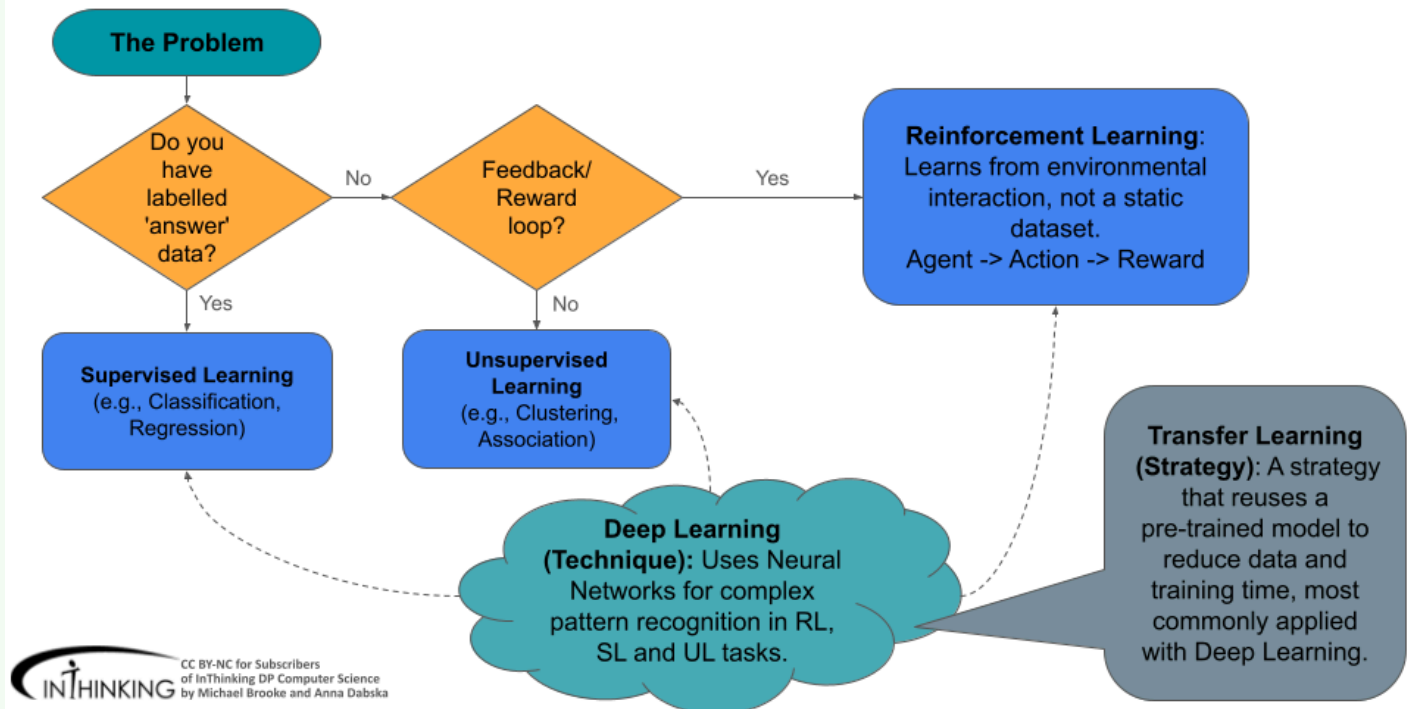
- **Medical imaging with limited data:** A model pre-trained on millions of general photographs already understands edges, textures, and shapes. Fine-tuning it on a few hundred specialist medical scans produces a capable diagnostic model without requiring thousands of rare examples.
- **Low-resource language translation:** A translation model pre-trained on large English and French datasets can be fine-tuned for a minority language with only a small parallel corpus, rather than requiring millions of translation pairs.
- **Object detection for specialist domains:** A developer fine-tunes a model pre-trained on general object detection to recognise specific industrial components, using only photographs they took themselves.

Common Exam Mistake Transfer learning is sometimes described as simply giving the model more data, or as a model that improves over time with use. Neither is correct. Transfer learning specifically means that a separately trained model, one trained on a different task or dataset, is used as the starting point. The knowledge transfers across tasks, not just across more of the same data.

Summary comparison

APPROACH	DATA / INPUT	TASK TYPES	APPLICATIONS	CATEGORY NOTE
Supervised Learning	Labelled data	Classification; Regression	<i>Sentiment analysis; medical imaging</i>	Type of learning
Unsupervised Learning	Unlabelled data	Clustering; Association	<i>Customer segmentation; market basket analysis</i>	Type of learning
Reinforcement Learning	No dataset -- learns from interaction	Policy optimisation	<i>Robotics navigation; game playing</i>	Type of learning
Deep Learning	Any (SL, UL, or RL tasks)	Complex pattern recognition in unstructured data	<i>Object detection; NLP; face recognition</i>	Technique, not a type
Transfer Learning	Small labelled dataset for the target task	Fine-tuning an existing model	<i>Rare disease detection; low-resource NLP</i>	Strategy, not a type

Relationships Between ML Approaches



A4.1.1 Quizlet.

Use the Quizlet to learn all the new key terms from this section. You can also choose different study modes to help you learn.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Quiz

Why not take a quick test on the content from this section to see how much you understood

1

A data scientist is given a large dataset of customer information and is tasked with finding natural groupings or segments within the customers without any pre-existing labels. Which ML approach is most suitable?

- A. ☐ Reinforcement Learning.
- B. ☐ Transfer Learning.
- C. ☐ Supervised Learning.
- D. ☒ Unsupervised Learning.

2

What is the primary characteristic of the data used to train a supervised learning model?

- A. ☐ It is very large.
- B. ☐ It has no labels.
- C. ☒ It is labelled with the correct outputs
- D. ☐ It is unstructured.

3

An AI is being trained to play a complex video game. It learns by receiving a score at the end of each game, which it tries to maximise over time. This is an example of:

- A. ☐ Unsupervised Learning.
- B. ☐ Supervised Learning.
- C. ☐ Transfer Learning.
- D. ☒ Reinforcement Learning.

4

Deep Learning is best described as:

- A. ☒ A technique using multi-layered neural networks, applicable to supervised, unsupervised
- B. ☐ A learning paradigm completely separate from supervised or unsupervised learning.
- C. ☐ A type of learning that only works with text data.
- D. ☐ A method for making models smaller and faster.



5

A developer wants to build an app that can identify different types of flowers. Instead of training a model from scratch, they download a powerful, pre-existing image recognition model and fine-tune it on their flower dataset. This strategy is known as:

- A. ☐ Reinforcement Learning.
- B. ☐ Unsupervised Learning.
- C. ☒ Transfer Learning.
- D. ☐ Supervised Learning.

6

The task of classifying emails as either 'Spam' or 'Not Spam' is a classic example of:

- A. ☐ Market basket analysis.
- B. ☐ An unsupervised clustering problem.
- C. ☒ A supervised classification problem.
- D. ☐ A reinforcement learning problem.

7

In reinforcement learning, what is the term for the learner or decision-maker that interacts with the world?

- A. ☐ The Policy
- B. ☒ The Agent
- C. ☐ The Environment
- D. ☐ The Reward

8

Which of the following applications is the best example of unsupervised learning?

- A. ☐ Identifying a person's face in a photograph.
- B. ☒ Finding that customers who buy bread also frequently buy milk.
- C. ☐ Training a robot to navigate a maze.
- D. ☐ Predicting a house's sale price based on its features.

9

The primary advantage of Transfer Learning is that it:

- A. ☒ Reduces the need for massive datasets and long training times.
- B. ☐ Can work with data that has no labels.
- C. ☐ Is the only way to work with image data.
- D. ☐ Always produces a more accurate model than one trained from scratch.

10

A model that processes medical scans to assist radiologists in diagnosing diseases is a real-world application of which technique?

- A. ☐ Robotics Navigation
- B. ☐ Sentiment Analysis
- C. ☒ Medical Imaging Diagnostics
- D. ☐ Market Basket Analysis

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